

ITEM SCREENING IN GRAPHICAL LOGLINEAR RASCH MODELS

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In behavioural sciences, local dependence and DIF are common, and purification procedures that eliminate items with these weaknesses often result in short scales with poor reliability. Graphical loglinear Rasch models (Kreiner & Christensen, in *Statistical Methods for Quality of Life Studies*, ed. by M. Mesbah, F.C. Cole & M.T. Lee, Kluwer Academic, pp. 187–203, 2002) where uniform DIF and uniform local dependence are permitted solve this dilemma by modelling the local dependence and DIF. Identifying loglinear Rasch models by a stepwise model search is often very time consuming, since the initial item analysis may disclose a great deal of spurious and misleading evidence of DIF and local dependence that has to be disposed of during the modelling procedure.

Like graphical models, graphical loglinear Rasch models possess Markov properties that are useful during the statistical analysis if they are used methodically. This paper describes how. It contains a systematic study of the Markov properties and the way they can be used to distinguish spurious from genuine evidence of DIF and local dependence and proposes a strategy for initial item screening that will reduce the time needed to identify a graphical loglinear Rasch model that fits the item responses. The last part of the paper illustrates the item screening procedure on simulated data and on data on the PF subscale measuring physical functioning in the SF36 Health Survey inventory.

Key words: chain graph models, graphical Rasch models, loglinear Rasch models, global Markov properties, differential item functioning, local dependence, Mantel–Haenszel analysis, partial gamma coefficient.

1. Introduction

In behavioral sciences, local dependence (LD) and differential item functioning (DIF) are common, and purification procedures (Lord, 1980; Park & Lautenschlager, 1990; French & Maller, 2007) that eliminate items suffering from these weaknesses often result in scales with few items and relatively poor reliability. In such situations, graphical loglinear Rasch models (GLLRM) (Kreiner & Christensen, 2002, 2004, 2006) may be useful, since they offer solutions to LD and DIF problems that do not require elimination of items.

Graphical Rasch models are latent structure models where measurement models with properties that are similar to those of ordinary Rasch models are embedded in multivariate structural frameworks consisting of chain graph models (Lauritzen, 1996). GLLRMs are extensions where uniform DIF (Mellenbergh, 1982; Hanson, 1998) and uniform LD are allowed. Like Rasch models, GLLRMs are unidimensional with sufficient person scores. Kreiner and Christensen (2006) and Kreiner (2007) claim that measurement by scores from such models is essentially valid and objective.

Graphical Rasch models and GLLRMs possess the same kind of global Markov properties as graphical models. This paper addresses inference in GLLRMs based on these properties. The purposes of the paper are fourfold: to provide a systematic account of the global Markov properties of GLLRMs and how they can be used during tests-of-fit of graphical Rasch models and GLLRMs, to elaborate on the risk that the analysis discloses spurious evidence of DIF and LD, to develop strategies and procedures for item screening that take care of the spurious evidence of DIF and LD without purification, and finally to illustrate the use of these procedures with real-world data.

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Spurious evidence against a statistical model is evidence that points at model errors that are non-existent. During item analysis, many different fit statistics (e.g., item fit statistics and DIF tests) are calculated. Each of these statistics tests a specific hypothesis derived from the IRT model. This entails a risk of misleading evidence since many hypotheses are false when a single assumption is violated.

In addition to modelling genuine departures from graphical Rasch models, a cautious GLRRM modelling procedure may also take care of the spurious evidence of DIF and LD. Inference in graphical Rasch models and GLLRMs is a two-step procedure. The first step checks that items satisfy requirements of Rasch models. The second step attempts to model DIF and LD by a stepwise model search procedure where interaction parameters representing DIF and LD are added to the graphical Rasch model. The stepwise model search in GLLRMs is similar to stepwise model search for loglinear models fitting multidimensional contingency tables, but the analysis is more complicated since it has to address high-dimensional contingency tables with many structural zeros. The methods developed for conditional inference in loglinear Rasch models (Kelderman, 1984, 1989, 1992) are useful, but are often challenged by slow convergence of the iterative procedures.

Model search procedures for loglinear models start from an initial model and proceed by adding or deleting either single interaction terms or blocks of interaction terms. Typical choices of initial models are (1) the main effects model, (2) the model containing all two-factor interactions or (3) the saturated model. The main factor GLLRM (the graphical Rasch model) is the natural starting point for a search for a GLLRM. A cautious stepwise model search procedure with this starting point that only adds or deletes a single interaction term in each step should in principle have no problems with spurious evidence of DIF and LD, but with many items and many instances of LD and DIF, this can be very time consuming. Starting from the saturated model or the two-factor interaction model suffers from the same problem to an even higher degree. Instead, a starting model containing interaction terms corresponding to the evidence of DIF and LD that was disclosed during the initial fit of the Rasch model may be better. However, comprehensive analyses of DIF and LD can produce a great deal of spurious evidence, and this may reduce the usefulness of this starting point.

The dataset analyzed in Section 7 is a case in point. It contains 10 items measuring physical function and five exogenous variables. The initial analysis disclosed evidence of LD for 22 pairs of items and DIF for seven items. In this case, model search for a GLLRM that fits the data is similar to loglinear model search in a 16-dimensional contingency table covering items, exogenous variables and the total score. Stepwise model search that moves forward from the Rasch model and frequently backtracks to check that interaction terms included at the start of the analysis are needed after inclusion of other interaction terms is possible, but very time consuming. Moving backwards from the saturated model or from the model containing all 95 two-factor interactions is even less attractive. Starting the model search procedure from the GLRRM containing the 29 interaction terms that match the evidence of DIF and LD is an obvious and better choice. If a large part of this evidence is spurious, this model is also far apart from the target model and will still be a challenge to the model fitting procedures.

The strategy for item screening described in this paper provides a solution to this dilemma. The purpose of the strategy is to identify some of the spurious evidence and to define a GLLRM based on the evidence that appears to be genuine. During our item screening, hypotheses are derived from the global Markov properties of graphical Rasch models and GLLRMs. For the graphical Rasch models, these include hypotheses of no DIF and hypotheses of local independence that can be tested by Maentel-Haenszel tests if items are dichotomous, or by partial γ coefficients (Davis, 1967; Agresti, 1984) if items are polytomous. In GLLRMs, the global Markov properties generate hypotheses that take acknowledged LD and DIF into account and where similar techniques can be used.

1.1. Notation and Abbreviations

Graphical Rasch models and GLLRMs contain four types of variables: (1) a continuous latent variable, Θ , (2) a set of scored responses to k items, $\mathbf{Y} = (Y_1, \dots, Y_k)$, (3) the score, $S = \sum_i Y_i$, and (4) exogenous variables (covariates), $\mathbf{X} = (X_1, \dots, X_m)$. Covariates may be criterion variables, known in advance to be associated with Θ . Throughout the paper we assume that $\text{Var}(\Theta) > 0$ and that all exogenous variables are discrete. The properties of graphical models on which the screening strategy depends also apply for continuous variables, but continuous variables require different measures of partial correlation and different tests of conditional independence. In addition to the total score, we also consider *rest-scores*, defined by subtracting items from the total score, for example, $R_a = S - Y_a$, $R_{ab} = S - Y_a - Y_b$, and *subscores*, $S_A = \sum_{i \in A} Y_i$ where $A \subset \{1, \dots, k\}$.

The notion of conditional independence is vital to the exposition in this paper. Following Dawid (1979) we denote conditional independence of A and B given C as $A \perp B | C$.

2. Chain Graph Models

Chain graph models are multivariate statistical models defined by a block-recursive structure and assumptions concerning conditional independence. The recursive structure partitions variables into, say, t recursive blocks, $V_1 \leftarrow V_2 \leftarrow \dots \leftarrow V_t$, often assumed to reflect temporal and/or causal structure. In the statistical model defined by this structure, the joint distribution of all variables is written as the product of the conditional probabilities of variables in specific blocks given all variables in prior blocks, $P(V) = \prod_{i=1 \dots t-1} P(V_i | V_{i+1}, \dots, V_t) P(V_t)$. In chain graph models, the probabilities are restricted by assuming that some pairs of variables are conditionally independent given all concurrent or prior variables. The model assumptions are encoded in chain graphs where variables are represented by nodes that are connected by lines or edges, unless the model specifically assumes that they are conditionally independent given all concurrent or prior variables. The edges in the graph are either undirected or directed. Edges connecting nodes in the same block are always undirected. Edges connecting nodes in different blocks are arrows pointing from prior to posterior blocks relative to the recursive structure. The graphs are called independence graphs since the independence assumptions of the model can be read directly off them, or Markov graphs because a number of Markov properties of the statistical model may be revealed by analyses of the graphs (Frank & Strauss, 1986; Lauritzen, 1996). We refer to Lauritzen (1996) for a complete account of these models, but have included a brief review of some of the vocabulary and theory of chain graph models in Appendix A for readers who are unfamiliar with the models. The key concepts of importance for this paper are separation in graphs, moral graphs, and *global* Markov properties (GMP).

GMP hypotheses are hypotheses of conditional independence that are derived from the global Markov properties of the models. To derive a GMP hypothesis from a chain graph one must first moralize the Markov graph by adding undirected edges between variables that are either parents to the same variable or parents to variables in the same chain components. Moralization also requires that arrows in the chain graph are replaced by undirected edges so that the moral graph is an undirected graph. GMP hypotheses can be generated for all pairs of variables that are unconnected in the moral graph. To define a conditioning set of variables for a GMP hypothesis we identify a separating subset of variables that intersect all paths between the two variables in the moral graph. The end result is a GMP hypothesis stating that the two variables are conditionally independent given the variables in the separating subset.

2.1. Functional Collapsibility

The notion of functional collapsibility in chain graph models is of particular interest to the theory of graphical Rasch models described in the next section. Let $\mathbf{X}$ and $\mathbf{Y}$ be vectors of random variables and assume that $P(\mathbf{X}, \mathbf{Y})$ is a chain graph model with Markov graph G_{XY} where all the Y -variables belong to the same recursive block. Also, let $S = f(\mathbf{Y})$ be a function of $\mathbf{Y}$. We say that the model is *functionally collapsible onto* $(\mathbf{X}, S)$ if $\mathbf{X} \perp \mathbf{Y} | S$. In this paper we focus on situations where $\mathbf{Y}$ is a vector of items, $\mathbf{X}$ is a vector of exogenous variables and S is the total score on all items. The definition of functional collapsibility and Theorem 1 below apply, however, to all types of variables and all types of functions.

Depending on f , the distribution of $(\mathbf{X}, S, \mathbf{Y})$ may have structural zeros because some of the outcomes of Y can have zero probability for specific outcomes of S . Despite this, Theorem 1 below shows that it is possible to construct a Markov graph, G_{XSY} so that $P(\mathbf{X}, S, \mathbf{Y})$ satisfies all the global Markov properties inherent in G_{XSY} .

Theorem 1. *If the distribution of $(\mathbf{X}, \mathbf{Y})$ is functionally collapsible onto $(\mathbf{X}, S)$, and if all variables of $\mathbf{Y}$ belong to the same recursive block, then the joint distribution of $(\mathbf{X}, S, \mathbf{Y})$ satisfies all the global Markov properties implied by the graph, G_{XSY} , defined in the following way.*

- (1) *The recursive structure of variables in G_{XSY} is the same as in G_{XY} with S in the same recursive block as $\mathbf{Y}$.*
- (2) *The set of nodes for $(\mathbf{Y}, S)$ is complete.*
- (3) *There are no edges connecting Y -variables to X -variables in G_{XSY} .*
- (4) *If there is no edge between X_i and X_j in G_{XY} , then there is no edge between X_i and X_j in G_{XSY} .*
- (5) *If there is no edge between X_i and Y_j for all $Y_j \in \mathbf{Y}$, then there is no edge between X_i and S in G_{XSY} .*

Proof: See Appendix A. □

Under functional collapsibility, Theorem 1 implies that G_{XSY} is a Markov graph for $P(\mathbf{X}, \mathbf{Y}, S)$ and that the subgraph G_{XS} defined by the nodes for $(\mathbf{X}, S)$ and the corresponding edges connecting these variables in G_{XSY} is a Markov graph for $P(\mathbf{X}, S)$. Finally, Corollary 2.1 that follows directly from Theorem 1 implies that there is no edge connecting X_i to S in G_{XS} if there are no edges connecting X_i to Y -variables in G_{XY} .

Corollary 2.1. *If $(\mathbf{X}, \mathbf{Y})$ is functionally collapsible onto $(\mathbf{X}, S)$, it follows that*

- (1) *If $X_i \perp X_j | (\mathbf{X}, \mathbf{Y}) \setminus (X_i, X_j)$ then $X_i \perp X_j | (\mathbf{X}, S) \setminus (X_i, X_j)$.*
- (2) *If $X_i \perp Y_j | (\mathbf{X}, \mathbf{Y}) \setminus (X_i, Y_j)$ for all $Y_j \in \mathbf{Y}$ then $X_i \perp S | \mathbf{X} \setminus X_i$.*

Functional collapsibility implies that a statistical analysis of the association among X -variables and Y -variables can be reduced to inference in the marginal $(\mathbf{X}, S)$ distribution. In this paper, $S = \sum_i Y_i$ and it leads to graphical Rasch models where G_{XSY} is referred to as a Rasch graph, but the definition of functional collapsibility extends to other functions and other models with or without latent variables.

3. Graphical Rasch Models

A graphical Rasch model is a chain graph model for $(\mathbf{Y}, \Theta, \mathbf{X})$ characterized by two Markov graphs, G_{IRT} and G_{Rasch} .

G_{IRT} is the conventional Markov graph associated with a chain graph model. The recursive block structure is $V_1 \leftarrow V_2 \leftarrow \dots \leftarrow V_t$ where $V_1, \dots, V_t$ is a sequence of disjoint subsets of variables so that $\bigcup_{i=1, \dots, t} V_i = \{\mathbf{Y}, \Theta, \mathbf{X}\}$. The model assumes that $\mathbf{Y} \subseteq V_1$ and $\Theta \in V_2$. The exogenous variables may be distributed across all or some of the blocks of the model.

Two sets of restrictions are imposed on G_{IRT} :

GRM1: $Y_i \perp Y_j | \Theta, \mathbf{Y} \setminus (Y_i, Y_j), \mathbf{X}$ for all pairs of items.

GRM2: $Y_i \perp X_j | \Theta, \mathbf{Y} \setminus Y_i, \mathbf{X} \setminus X_j$ for pairs of items and exogenous variables.

These two restrictions imply that no edges connect items to each other or to exogenous variables in G_{IRT} . The structure of this model entails that the distribution of $(\mathbf{Y}, \Theta, \mathbf{X})$ can be decomposed as a product of a measurement component describing the dependence of items on Θ and a structural component consisting of the marginal distribution of $(\Theta, \mathbf{X})$, $P((\mathbf{Y}, \Theta, \mathbf{X})) = P(\mathbf{Y} | \Theta) P(\Theta, \mathbf{X})$ where $P(\Theta, \mathbf{X})$ is a chain graph model defined by the subgraph of G_{IRT} containing Θ and $\mathbf{X}$ but not $\mathbf{Y}$. GRM1 and GRM2 are assumptions of local independence (no LD) and no DIF. In the theory of chain graph models, these assumptions are referred to as pairwise block-recursive Markov properties (Lauritzen, 1996, p. 54). From these properties and Theorem 3.36 of Lauritzen (1996, p. 59), it follows that $P(\mathbf{Y} | \Theta) = \prod_i P(Y_i | \Theta)$ and $\mathbf{Y} \perp \mathbf{X} | \Theta$ and, since obviously the opposite is also true, GRM1 and GRM2 are equivalent to conventional definitions of conditional independence and no DIF. We also assume that the relationships between items and Θ are monotonic in the sense that the expected item score $P(Y_i \geq y | \Theta = \theta)$ is a nondecreasing function of θ for all y . From this it follows that the measurement component of a graphical Rasch model satisfies all requirements of construct related criterion validity (Rosenbaum, 1989) and all requirements of conventional IRT models (see Holland & Hoskens, 2003, who include no DIF as one of the IRT requirements). For this reason, G_{IRT} is referred to as an IRT graph. Consequences of the assumption of monotonicity are discussed by Van der Ark and Bergsma (2010).

G_{Rasch} adds the total score to the block with the items, replaces the arrows pointing from Θ to items with one arrow connecting Θ to S , and adds edges connecting all items to each other. G_{Rasch} is motivated by the assumption that the set of items is functionally collapsible onto the score in the model defined by G_{IRT} :

GRM3: $\mathbf{Y} \perp \Theta, \mathbf{X} | S$.

From GRM3 it follows that the distribution of $(\mathbf{Y}, S, \Theta, \mathbf{X})$ satisfies all the global Markov properties entailed by G_{Rasch} . $S = \sum_i Y_i$, Y_a is completely determined by S and the outcomes on all other variables. For this reason, some of the pairwise Markov properties implied by G_{Rasch} , for example, $Y_a \perp X_b | \Theta, S, \mathbf{Y} \setminus Y_a, \mathbf{X} \setminus X_b$, are trivially true and cannot be regarded as restrictions on the $(\mathbf{Y}, S, \Theta, \mathbf{X})$ distribution. It nevertheless follows from Theorem 1 that the distribution satisfies all other Markov properties implied by G_{Rasch} . For this reason, we claim that GRM1–GRM3 together define the chain graph model for $(\mathbf{Y}, S, \Theta, \mathbf{X})$ characterized by G_{Rasch} . In this model, S intersects all paths from items to Θ and $\mathbf{X}$. In addition to the no DIF implied by the G_{IRT} , there will therefore be no DIF among items conditionally given the total score since $\mathbf{Y} \perp \mathbf{X} | S$.

Figures 1a and 1b show the Markov graphs of a graphical Rasch model with four items and four exogenous variables. In G_{Rasch} , edges connecting items are dotted to remind the reader that the conditional dependence between the items is artificial (induced by conditioning on S). In terms of the mathematical properties of the G_{Rasch} , there is no difference between dotted and solid lines.

G_{Rasch} implies Bayesian sufficiency in the sense that $P(\mathbf{Y} | \Theta = \theta) = P(\mathbf{Y} | S) P(S | \Theta = \theta)$ since $\mathbf{Y} \perp \Theta | S$. From this together with GRM1, GRM2, and $\text{Var}(\Theta) > 0$ it follows that $P(\mathbf{Y} | \Theta = \theta, \mathbf{X})$ is the distribution of item responses from a Rasch model (Andersen, 1977;

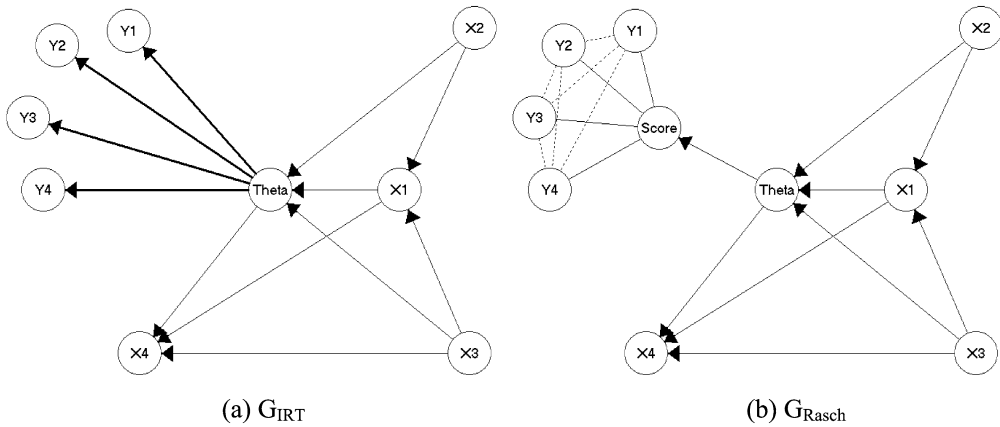


FIGURE 1.

IRT and Rasch graphs for four items, one criterion variable and three covariates. *Bold lines* correspond to known monotonic relationships. The *dashed edges* between the items in the Rasch graph are included, since items cannot be conditionally independent given the total score.

TABLE 1.
Manifest relationships in graphical Rasch models.

Type of association	Name	Definition
Positive correlation	M1	Y_i and Y_j are positively correlated for all pairs of items
	M2	Y_a is positively monotonically related to the rest-score R_a and all subscores S_B where $Y_a \notin B$
	M3	If X is positively related to Θ , then X will also be positively related to S , to all subscores, S_A , and all item responses Y_i
Positive partial correlation	S1	S is <i>conditionally</i> positively associated with all criterion variables
Conditional independence	C1	$(Y_1, \dots, Y_k) \perp (X_1, \dots, X_m) \mid S$
	C2	$Y_i \perp X_j \mid S$ for all $i = 1 \dots k$ and $j = 1 \dots m$
	C3	$Y_a \perp Y_b \mid S_A$ and $Y_a \perp Y_b \mid S_B$ for all $Y_a \in A$ and $Y_b \in B$
	C4	$Y_a \perp Y_b \mid R_a$ and $Y_a \perp Y_b \mid R_b$
	S2	$S \perp X \mid Z$ if Z is a subset of exogenous variables so that $\Theta \perp X \mid Z$

Note: M1 and M2 require that $\text{Var}(\Theta) > 0$.

Fischer, 1995):

$$P(\mathbf{Y} = \mathbf{y} \mid \Theta = \theta, \mathbf{X} = \mathbf{x}) = \exp\left(\alpha_0 + \sum_{i=1}^k (\theta y_i + \alpha_{iy_i})\right) = \exp\left(\alpha_0 + s\theta + \sum_{i=1}^k \alpha_{iy_i}\right). \quad (1)$$

We refer to Tjur (1982) for a discussion of sufficiency in Rasch models for dichotomous items with random person effects. Our treatment of Rasch models as graphical models shows that Tjur's results extend to models with polytomous items and exogenous variables.

Table 1 summarizes the relationships among the manifest variables in graphical Rasch models where $\text{Var}(\Theta) > 0$. M1, M2, M3, and S1 are consequences of GRM1 and GRM2 (Rosenbaum, 1984; Holland & Rosenbaum, 1986; Junker, 1993; Junker & Sijtsma, 2000) and apply to all IRT models defined by G_{IRT} . Item responses are said to be consistent if M1 and M2 apply. M3 provides the background for analysis of criterion validity. Following Rosenbaum (1989), one may argue that criterion validity is a necessary consequence of construct validity. C1 and C2 are GMP hypotheses derived from G_{Rasch} by marginalizing G_{Rasch} over the latent variable

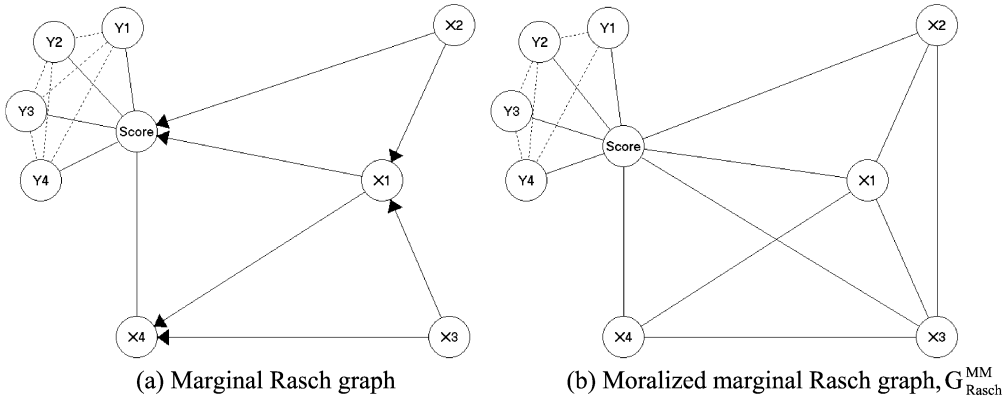


FIGURE 2.

(a) The marginal Rasch Graph and (b) the moralized marginal Rasch graph, $G_{\text{Rasch}}^{\text{MM}}$. Edges between items are drawn as *dashed lined* since items are conditionally dependent given the total score.

(Figure 2a) and then moralizing the marginal chain graph containing the manifest variables (Figure 2b). The moralized marginal Rasch graph is referred to as $G_{\text{Rasch}}^{\text{MM}}$. Since S separates items from exogenous variables in $G_{\text{Rasch}}^{\text{MM}}$, it follows that C1 and C2 are true under the graphical Rasch model. C2 motivates Mantel–Haenszel tests of no DIF for dichotomous items and exogenous variables (Holland & Thayer, 1988). C3 and C4 are GMP hypotheses that follow from moralization of Rasch graphs for subscales. Tjur (1982) showed C4 for three dichotomous items. The graphical structure of Rasch models implies that Tjur’s result extends to any number of dichotomous or polytomous items. Finally, S2 is also a consequence of the global Markov properties of G_{Rasch} .

4. Graphical Loglinear Rasch Models

Loglinear Rasch models (Kelderman 1984, 1989, 1992, 2005) are well-known extensions of the Rasch model. A graphical loglinear Rasch model (GLLRM) is a loglinear Rasch model embedded in a chain graph structure.

GLLRMs add interaction terms between items and between items and exogenous variables to the Rasch models. It is convenient to distinguish between three types of loglinear generators. One type is a set of DIF generators $D = (D_1, \dots, D_{nD})$ where $D_i = (A_i, Z_i)$ with $A_i \in \{Y_1, \dots, Y_k\}$ and $Z_i \in \{X_1, \dots, X_m\}$. Z_i is called a source of DIF of A_i . A second type is a set of LD generators, $L = (L_1, \dots, L_{nL})$ consisting of pairs of items $L_i = (U_i, V_i)$ where $\{U_i, V_i\} \subset \{Y_1, \dots, Y_k\}$. Finally, the third type consists of higher order interactions, $G = (G_1, \dots, G_{nG})$, where $G_i \subset \{Y_1, \dots, Y_k, X_1, \dots, X_m\}$ contains at least one item, which describe more complex DIF and LD structure. With these definitions, the conditional distribution of item responses given Θ and exogenous variables is

$$\begin{aligned}
 P(\mathbf{Y} = \mathbf{y} | \Theta = \theta, \mathbf{X} = \mathbf{x}) \\
 &= \exp \left(\alpha_0 + \sum_{i=1}^k (\theta y_i + \alpha_{iy_i}) + \sum_{i=1}^{nD} \delta_i(a_i, z_i) + \sum_{i=1}^{nL} \lambda_i(u_i, v_i) + \sum_{i=1}^{nG} \mu_i(g_i) \right) \\
 &= \exp \left(\alpha_0 + s\theta + \sum_{i=1}^k \alpha_{iy_i} + \sum_{i=1}^{nD} \delta_i(a_i, z_i) + \sum_{i=1}^{nL} \lambda_i(u_i, v_i) + \sum_{i=1}^{nG} \mu_i(g_i) \right), \quad (2)
 \end{aligned}$$

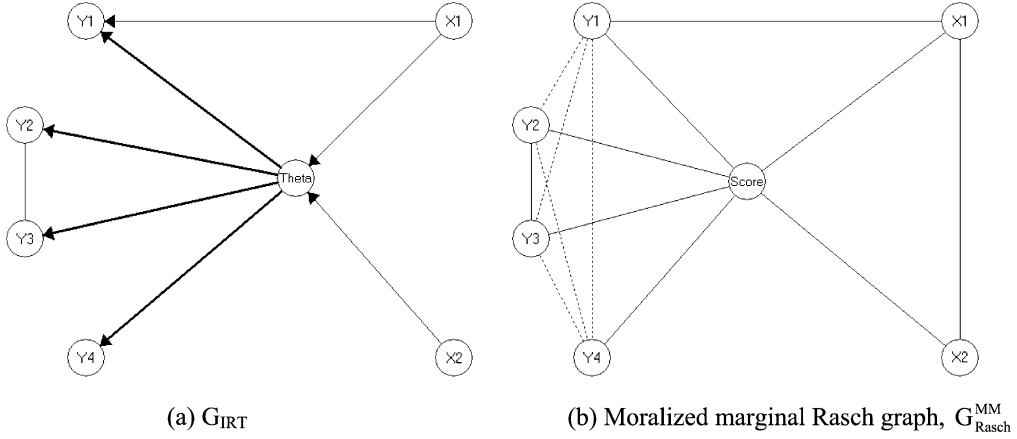


FIGURE 3.

IRT graphs derived from a GLLRM with two locally dependent items and one DIF item. *Dashed edges* refer to conditional association induced by conditioning on the total score on all items.

where S is sufficient for θ , and DIF and LD are uniform (Hanson, 1998) since higher order interactions in the model are independent of θ (see Hoskens & De Boeck, 1997, and Ip, 2002, for discussions of IRT models where LD depends on θ).

In (2) items may have several sources of DIF. The set of DIF sources of Y_i is called $\text{Source}(Y_i) \subseteq \{X_1, \dots, X_m\}$. Likewise, an exogenous variable can be the source of DIF for several items. The set of items for which X_j is a DIF source is called $\text{DIF}(X_j) \subseteq \{Y_1, \dots, Y_k\}$.

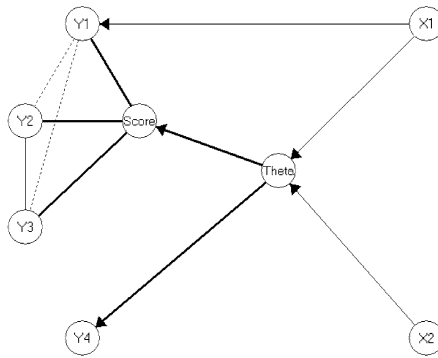
In GLLRMs, Y_i and Y_j are locally independent ($Y_i \perp Y_j | \Theta, \mathbf{Y} \setminus \{Y_i, Y_j\}, \mathbf{X}$) and X_j is not a source of DIF for Y_i ($Y_i \perp X_j | \Theta, \mathbf{Y} \setminus Y_i, \mathbf{X} \setminus X_j$) if and only if the corresponding interaction parameters are equal to 0. Since the assumptions of conditional independence are GRM1 and GRM2 assumptions, it follows that GLLRMs are models defined by *subsets* of the requirements of graphical Rasch models. Also, because S is sufficient, GLLRMs also have two Markov graphs. G_{IRT} includes undirected edges between LD items and edges or arrows connecting DIF items to DIF sources. The chain components of G_{IRT} containing subsets of items connected by paths are called the item components of the GLLRM. Item components are locally independent, and total score on all items of an item component is distributed as a partial credit item (Kreiner & Christensen, 2006). G_{Rasch} has the same edges and arrows connecting DIF items to DIF sources as G_{IRT} .

4.1. Global Markov Properties and GMP Hypotheses

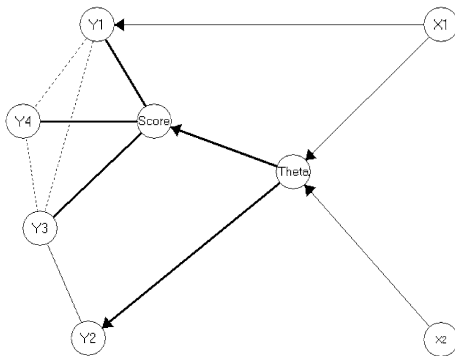
Like graphical Rasch models, GLLRMs also have global Markov properties. Example 1 illustrates the differences between the global Markov properties of graphical Rasch models and the global Markov properties of GLLRMs.

Example 1. The GLLRM in Figure 3a has three item components: $\{Y_1\}$, $\{Y_2, Y_3\}$ and $\{Y_4\}$ and one DIF item relative to X_1 . To identify the global Markov properties of this model, we moralize G_{Rasch} after marginalization over Θ . The result, G_{Rasch}^{MM} , is shown in Figure 3b.

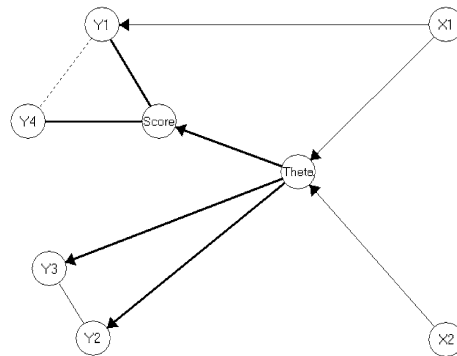
In G_{Rasch}^{MM} , Y_1 and X_2 are connected by two paths: $(X_2 - S - Y_1)$ and $(X_2 - X_1 - Y_1)$, and S and X_1 are both needed to separate Y_1 from X_2 . From this it follows that the minimal GMP hypothesis for a test of $X_2 \notin \text{Source}(Y_1)$ is $Y_1 \perp X_2 | S, X_1$. The C2 hypothesis $Y_1 \perp X_2 | S$ and all the other C2 hypotheses under the graphical Rasch model are not necessarily true under the GLLRM, since S



(a) Rasch graph with Y_4 as an exogenous variable. In this graph the score is $R_4 = Y_1 + Y_2 + Y_3$



(b) Rasch graph with Y_3 as an exogenous variable. The score is $R_2 = Y_1 + Y_3 + Y_4$



(c) Rasch graph with (Y_2, Y_3) as exogenous variables. The score is $R_{23} = Y_1 + Y_4$

FIGURE 4.

IRT graphs for rest-scores in the GLLRM defined in (a). *Dashed edges* are included due to conditioning with the scores.

does not interrupt all the paths between items and exogenous variables in G_{Rasch}^{MM} . A test of a C2 hypothesis may therefore result in spurious evidence of DIF.

The remaining set of separation properties and corresponding minimal GMP hypotheses for a test of no DIF are read from G_{Rasch}^{MM} . They are $Y_i \perp X_2 | S, X_1$ and $Y_i \perp X_2 | S, Y_1$, and $Y_i \perp X_1 | S, Y_1$.

Figure 4 illustrates the derivation of GMP hypotheses for tests of local independence of Y_2 and Y_4 under the GLLRM in Figure 3. The first step is to redefine the LD problem as a DIF problem by considering an item as an exogenous variable. To keep items in the model as exogenous variables, we subtract the items' scores from the total score and proceed to derive the minimal GMP hypotheses as above. In Figure 4a, $Y_2 \perp Y_4 | R_4$ is a minimal GMP hypothesis. This hypothesis is one of the C4 hypotheses of local independence under the graphical Rasch model. In Figure 4c, the score is equal to the rest-score without the two LD items and $Y_2 \perp Y_4 | R_{23}$ is another minimal GMP hypothesis. In Figure 4b, the minimal GMP hypotheses is $Y_2 \perp Y_4 | R_2, Y_3$. Since $R_2 = Y_1 + Y_3 + Y_4$, this hypothesis is the same as $Y_2 \perp Y_4 | R_{23}, Y_3$. This hypothesis is also a GMP hypothesis under the model defined by Figure 4c, but it is not minimal and therefore not counted as a minimal GMP hypothesis under the model defined by Figure 3. Concluding this example, we finally note that the other C4 hypothesis derived from the Rasch model, $Y_2 \perp Y_4 | R_2$, is not a GMP hypothesis and therefore not automatically true. Testing this hypothesis as part of a test of the Rasch model may again lead to spurious evidence of LD between Y_2 and Y_4 .

In situations with many items and a complicated structure, specialized software is required to identify the minimal GMP hypotheses. Instead, the following GMP hypotheses of no DIF and no LD may be used. Let Y_{DIF} be the set of DIF items and X_{Source} be the set of DIF sources. Four different types of paths connect X_j to Y_i in G_{Rasch}^{MM} : (1) $(X_j - S - Y_i)$, (2) $(X_j \cdots X_k \in \text{Source}(Y_i) - Y_i)$, (3) $(X_j - Y_a \in \text{DIF}(X_j) \setminus Y_i - Y_i)$, and (4) $(X_j \cdots X_k \in X_{\text{Source}} \setminus X_j - Y_b \in \text{DIF}(X_k) \setminus Y_i \subseteq Y_{\text{DIF}} \setminus Y_i - Y_i)$. The conditioning set for the no DIF hypothesis must include variables that intersect all these paths. We include S to intersect the first path, $\text{Source}(Y_i)$ to intersect the second type, $\text{DIF}(X_j)$ to intersect the third type and *either* $X_{\text{Source}} \setminus X_j$ or $Y_{\text{DIF}} \setminus Y_i$ to intersect the fourth type. Since $\text{Source}(Y_i) \subseteq X_{\text{Source}}$ and $\text{DIF}(X_j) \subseteq Y_{\text{DIF}}$, the GMP hypotheses defined by these separating sets are

$$\mathbf{C5}: Y_i \perp X_j | S, Y_{\text{DIF}} \setminus Y_i, \text{Source}(Y_i) \quad \mathbf{C6}: Y_i \perp X_j | S, X_{\text{Source}} \setminus X_j, \text{DIF}(X_j)$$

for all $i = 1 \dots k$ and $j = 1 \dots m$.

Similar arguments may be used to derive GMP hypotheses for tests of LD between Y_a and Y_b in a GLLRM. Let Y_A and Y_B be two locally independent subsets of items with corresponding subscores S_A or S_B where $Y_a \in Y_A$ and $Y_b \in Y_B$. If there is no DIF in Y_A a test of $Y_a \perp Y_b | S_A$ is a valid LD test. If some items in Y_A function differentially, the local chain Markov properties (see Appendix A) require that the hypothesis should be replaced by $Y_a \perp Y_b | S_A, Y_C$ and/or $Y_a \perp Y_b | S_A, X_C$ where Y_C is the subset of DIF items in Y_A and X_C is the set of DIF sources for Y_C . Similar arguments pertain to Y_B where Y_D is the subset of DIF items in Y_B and X_D are the DIF sources. If Y_a and Y_b are locally independent and if Y_A and Y_B are defined as above, it follows that the following four hypotheses hold:

$$\begin{aligned} \mathbf{C7}: Y_a \perp Y_b | S_A, Y_C, & \quad \mathbf{C8}: Y_a \perp Y_b | S_A, X_C, \\ \mathbf{C9}: Y_a \perp Y_b | S_B, Y_D, & \quad \mathbf{C10}: Y_a \perp Y_b | S_B, X_D. \end{aligned}$$

C7 and C9 apply for all partitioning of items into two locally independent subsets, except when $Y_C = Y_A$ or $Y_D = Y_B$. C8 and C10 always apply. It is, of course, impractical to consider all partitions of items satisfying C7–C10. In practical situations, we therefore only consider the situations where Y_A is the item component with Y_a , and Y_B is the remaining items $Y_B = Y \setminus \{Y_a, Y_b, Y_A\}$; or where Y_B is the item component with Y_b , and $Y_A = Y \setminus \{Y_a, Y_b, Y_B\}$.

4.2. Graphical Loglinear Rasch Modeling

Given a specific outcome on Θ , $P(\mathbf{Y}, \mathbf{X} | \Theta = \theta) = P(\mathbf{Y} | \mathbf{X}, \Theta = \theta) P(\mathbf{X} | \Theta = \theta)$ is an ordinary loglinear model for the multidimensional table counting responses to item and exogenous variables where item main effects and interaction parameters relating to exogenous variables may depend on θ and where interaction parameters referring to items are independent of θ .

GLLRM modelling attempts to identify which GLLRM provides the best fit to data. If there were no latent variable, it would be a standard exercise in stepwise loglinear model search. Instead, GLLRM modelling attempts to identify the loglinear model fitting the multidimensional contingency table counting responses to items, exogenous variables and the sufficient person score. The loglinear model is a quasi loglinear model, since the table containing item responses together with the total score is incomplete (Bishop, Fienberg, & Holland, 1975). Kelderman (1984, 1992) has solved the technical problems of fitting such models. Except for the use of the techniques, model search among GLLRMs is similar to standard loglinear model search: starting from an appropriate initial model, interaction terms are either added to or deleted from the model until an adequate fit has been obtained (see Bartolucci & Forcina, 2005, for a similar approach to analysis by loglinear non-parametric IRT models).

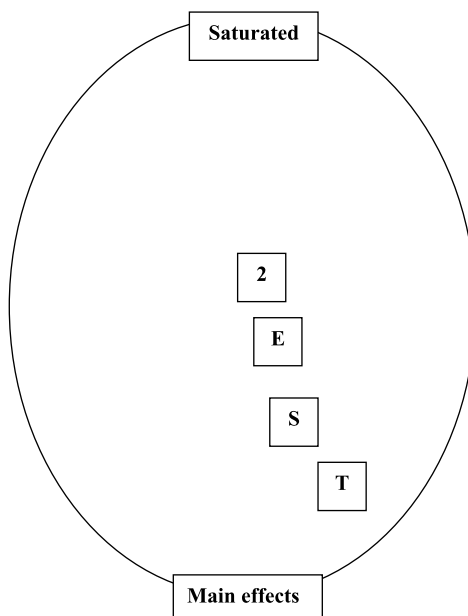


FIGURE 5.

The GLLRM model space. Six models are shown: the saturated model, the main effects model, 2: the model containing all two-factor interactions, *E*: the model defined by all evidence of DIF and LD, *S*: The model defined by item screening, and *T*: the target model.

The choice of the initial model is very important in terms of computing time. The farther away the initial model lies from the target model, the longer the search time. Common model search procedures for conventional loglinear models either move forward from the main effects model without interactions or backwards from the saturated model containing all possible interaction terms. Other options could be to start from the model containing all two-factor interaction terms or from the graphical loglinear model defined by the screening procedure of Kreiner (1986).

Figure 5 is an idealized illustration showing the positions of six different models with the saturated model at the top and the main effects model (the graphical Rasch model) at the bottom. The T-Model is the target model providing the best fit to data; the needle in the haystack that we want to find in as short a time as possible. This model is posited closer to the main effects model than to the saturated model because the items were constructed to meet the requirements of the main effects model. The other three models are models that may be used as starting points for the model search instead of the main effects model or the saturated model. The 2-model contains all 2-factor interaction model. The E-model contains all 2-factor interactions supported by the evidence of DIF and LD that was disclosed during the fit of the Rasch model. The discussion of Example 1 made the points that some of the evidence of DIF and LD may be spurious. The S-model in Figure 5 is defined by the interaction terms supported by genuine evidence. The S-model is equal to the target model if the true model actually is a GLLRM and if there were neither Type I or Type II errors during the fit of the Rasch model. Such good luck cannot be expected, but we do expect the S-model to be closer to the target model than all the other models in Figure 5. Using this model as a starting point will therefore reduce the search for the T-model to a minimum.

5. Testing GMP Hypotheses: Technical Problems and Spurious Evidence

5.1. Technical Issues

In many cases, tests of GMP hypotheses of DIF and LD are tests of conditional independence in large sparse contingency tables. For such tests to be feasible, three technical problems have to be dealt with; namely the choice of test statistics, the assessment of significance in large sparse tables, and the re-evaluation of significance in light of the large number of tests being performed during the analysis.

5.1.1. Test Statistics for Hypotheses of no DIF The psychometric literature contains a large number of tests of no DIF in Rasch and other IRT models (see Penfield & Camilli, 2007, for a relatively recent review and additional references). It is beyond the scope of this paper to give a complete review of all these proposals. Here, we use fit statistics that are natural under conditional inference in Rasch models where associations are analyzed conditionally given the total score. The Mantel–Haenszel test of no DIF for dichotomous items is one of these methods. It was proposed by Holland and Thayer (1988) who noticed that the hypothesis $Y_i \perp X_j | S$ “holds exactly” (our italics) under the Rasch model. It follows also from Holland and Thayer that log odds-ratios measuring the association of Y_i and X_j in the $Y_i \times X_j \times S$ table are constant across the different levels of S and that the estimate of the common log odds-ratio is an estimate of the DIF interaction parameter in (2) if DIF is uniform.

The majority of studies examining the performance of the Mantel–Haenszel procedure for analysis of DIF do so under 2- or 3-parameter IRT models (e.g., Mazor, Clauser, & Hambleton, 1992; Ackerman, 1992; Raju, Drasgow, & Slinde, 1993; Clauser, Mazor, & Hambleton, 1994; Fidalgo, Mellenbergh, & Muniz, 2000; Penfield, 2001; Sue & Wang, 2005; Finch, 2005; Williams & Beretvas, 2006), despite the fact that the hypothesis of conditional independence of items and exogenous variables given the total score is not “exactly” true under these models. In large sample studies with few items, there is a considerable risk that the Mantel–Haenszel test of no DIF will disclose spurious evidence of DIF for exactly this reason.

Other analyses in the same vein include the loglinear analysis of Mellenbergh (1982), the logistic regression analysis of Swaminathan & Rogers (1990) and the loglinear Rasch model test of Kelderman (1984, p. 234, Table 3, test h). All of these methods are valid under the Rasch model and provide estimates of the same conditional odds-ratio if there is but only one item with uniform DIF. It is worth noting that the Rasch model requires that the score is included without categorization, since the Rasch model does not guarantee that items and exogenous variables are conditionally independent given score groups defined by score intervals. For the same reason, the logistic regression analysis also requires that the score is treated as a categorical variable (see Swaminathan and Rogers 1990, for a discussion of the connection between the Rasch model and the logistic regression analysis of DIF).

For polytomous items we have to use generalizations of these methods that accommodate ordinal items and exogenous variables with more than two outcomes. In this paper, we use the partial γ coefficient, but other possibilities also exist (Kelderman, 1989; Zumbo, 1999; Sue & Wang, 2005).

5.1.2. Test Statistics for Hypotheses of Local Independence It follows from the representation of the Rasch model as a graphical model that all tests for no DIF also apply for LD analysis by redefinition of some of the items as exogenous variables followed by analysis of DIF in the rest-score relative to these items. For dichotomous items, this means that Mantel–Haenszel methods and partial γ coefficients apply in exactly the same way as for DIF analyses. The ground breaking papers for LD analyses of this kind are the papers by Holland (1981), Rosenbaum (1984), Holland and Rosenbaum (1986), Rosenbaum (1988) and Ip (2001). In most of these

cases, the analysis of the conditional association of two items is carried out by conditioning on either the total score on all items or by conditioning on the rest-score without the items. The result that a test of local independence in Rasch models should condition on rest-scores without one of two items of interest was first derived by Kreiner (1993/2006).

5.1.3. Assessment of Significance The Monte Carlo procedure described by Kreiner (1987) provides unbiased estimates of exact conditional p -values. Besag and Clifford (1991) describe a simple procedure for sequential Monte Carlo tests that minimizes computing time by stopping the Monte Carlo process as soon as it is clear that the test result will be regarded as insignificant at a specific critical level. Christensen and Kreiner (2007) describe a Repeated Monte Carlo procedure that interrupts the process when it is highly unlikely that the test result will turn out to be significant. Finally, the multiple testing problems can be dealt with by the Benjamini–Hochberg (1995) procedure that controls the false discovery rate at a specific level. An illustration of how this works for tests of local independence can be found in Ip (2001).

5.2. Spurious Evidence

A rejected GMP hypothesis is evidence of LD or DIF, but other departures from the GLLRM increase the risk of spurious evidence. The assumption that the total score separates item responses from Θ in G_{Rasch} is not met if items fit a 2-parameter IRT model, because then G_{Rasch} has arrows connecting Θ to items. Items and exogenous variables are therefore not conditionally independent, and GMP hypotheses of no DIF may be rejected even when there is no DIF at all. Similar arguments apply for the LD tests: the C2 and C4 hypotheses are false under the 2-parameter model, and spurious evidence of DIF and LD may turn up.

Distinguishing between true and spurious evidence of DIF within a GLLRM is in principle simple. If the tests of GMP hypotheses suggest that several items function differentially relative to one or more DIF sources, and if the analysis discloses evidence of LD among several sets of items, the next step is to construct a GLLRM that takes all the evidence into account. A backward elimination procedure eliminating insignificant interactions from this model by tests of GMP hypotheses will take care of the spurious evidence of DIF and LD. Keeping track of and updating the global Markov properties after each step is, however, not easy. Instead, the simpler item screening strategy described in the next section may be preferable.

5.3. Examples of Spurious Evidence

To illustrate the risk of spurious evidence we analyze three simulated data sets with five dichotomous items (labeled A to E) and two binary exogenous variables (labeled X and Z) that meet all assumptions except one. All examples are large sample studies ($n = 10,000$) set up so that spurious evidence is bound to turn up.

Example 2 (One DIF Item with One DIF Source). In this example, item B functions differentially relative to X . DIF is uniform with $\delta_{BX} = 0.6931$. The expected Mantel–Haenszel estimate of DIF is equal to 2 corresponding to an expected partial γ coefficient equal to 0.333.

Table 2 shows the partial γ coefficients calculated during analysis of DIF for all items relative to X and Z . The hypothesis of no DIF is rejected for A relative to X , and the observed γ is very close to the expected value. The hypotheses of no DIF are, however, also rejected for other pairs of items and exogenous variables.

Furthermore, analyses of LD by tests of C4 hypotheses reject local independence in three cases: $A \perp B | R_A : \gamma = 0.110, p = 0.003$; $B \perp D | R_D : \gamma = 0.071, p = 0.046$; $A \perp E | R_E : \gamma = -0.110, p = 0.011$. All this evidence is spurious, the reason being that $P(B = 1 | \Theta = \theta) =$

TABLE 2.
Example 2: γ coefficients for tests of C2 hypotheses of no DIF.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
<i>X</i>	−0.092**	0.332***	−0.061*	−0.043	−0.138***
<i>Z</i>	−0.119***	−0.089**	−0.009	0.009	0.014

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

TABLE 3.
Example 3: γ coefficients for tests of C4 hypotheses of local independence given rest-scores without the row item.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
<i>A</i>		−0.055	0.031	0.036	−0.025
<i>B</i>	−0.185***		0.347***	−0.182***	−0.109*
<i>C</i>	−0.128***	0.311***		−0.130***	−0.114**
<i>D</i>	0.037	−0.035	0.008		−0.009
<i>E</i>	−0.037	0.001	−0.010	0.030	

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

$\sum_x P(B = 1|\Theta = \theta, X = x)P(X = x|\Theta = \theta)$ is not a Rasch probability. Since B is included in rest-scores R_A , R_D , and R_E , it follows that these rest-scores are not sufficient for θ and therefore that the C4 hypotheses are false.

Example 3 (Two LD Items). In this example, items B and C are uniformly LD with $\lambda_{BC} = 0.6933$. The expected γ measuring the conditional association during the analysis of local dependence is again equal to 0.333. Table 3 shows the partial γ coefficients calculated during the analysis of local dependence by tests of C4 hypotheses.

The observed γ 's measuring the conditional association of items B and C are close to the expected γ , but the remaining evidence of LD is spurious.

Example 4 (One Item with too Strong Item Discrimination). In this example, there is no DIF and no LD, but the item discrimination of item D is 3 rather than 1 as required by the Rasch model. Table 4 summarizes test results relating to LD and DIF.

The evidence in Table 4 correctly rejects the Rasch model, but draws a completely misleading picture of why the model does not fit. The reason for the spurious evidence is that the score is not sufficient, and thus *all* the hypotheses tested by Mantel–Haenszel tests and partial γ coefficients are false.

5.4. Comments

Perusing the results of Examples 2 and 3 it is perhaps not difficult to guess what the real problem is, but the point is nevertheless made. Spurious evidence did turn up. Example 4 is a much more challenging example illustrating first that all evidence of DIF and LD can be spurious

TABLE 4.
Example 4: γ coefficients for tests of C4 and C2.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
Local dependence					
<i>A</i>		−0.088 [*]	−0.123 ^{***}	0.449 ^{***}	−0.122
<i>B</i>	−0.087 [*]		−0.189 ^{***}	0.426 ^{***}	−0.064
<i>C</i>	−0.090 [*]	−0.174 ^{***}		0.440 ^{***}	−0.168
<i>D</i>	0.016	−0.003	0.018		−0.034
<i>E</i>	−0.150 ^{***}	−0.062	−0.172 ^{***}	0.397	
DIF					
<i>X</i>	−0.045	−0.064 [*]	−0.103 ^{***}	0.398 ^{***}	−0.134 ^{***}
<i>Z</i>	−0.131 ^{***}	−0.050	−0.142 ^{***}	0.514 ^{***}	−0.171 ^{***}

^{*} $p < 0.05$.
^{**} $p < 0.01$.
^{***} $p < 0.001$.

TABLE 5.
A strategy for item screening.

Step 1	Analysis of consistency	M1, M2 and M3
Step 2	Analysis of DIF and LD	C2 and C4
Step 3	(a) Elimination of spurious evidence of LD	
	(b) Elimination of spurious evidence for items with more than one DIF source by tests of $Y_i \perp X_j S$, $\text{Source}(Y_i)$	
	(c) Elimination of spurious evidence for sources with an apparent DIF effect on more than one item by tests of $Y_i \perp X_j S$, $\text{DIF}(X_j)$	
Step 4	Analysis of the association between the score and exogenous covariates	S1 and S2
Step 5	Definition of a GLLRM and analysis of the global Markov properties of G_{Rasch}	
Step 6	Tests of global Markov properties of the GLLRM	C5, C7 and C9

and second, that it can be difficult to recognize the precise point of departure from the Rasch model if the focus on potential model problems is too narrow. We return to these examples in Section 6.

6. Item Screening

Item screening consists of the six steps that are summarized in Table 5.

Step 1 assesses the consistency of item responses by analyses of the marginal associations among items, among items and rest-scores, among items and exogenous variables and among the total score and exogenous variables. This is natural during initialization of the item screening procedure and may lead to purification before the GLLRM modelling starts.

Step 2 tests C2 and C4 hypotheses of no LD and no DIF in Rasch models. Assessment of significance of the complete set of tests is adjusted by the Benjamini-Hochberg procedure to control the false discovery rate at 5%. The result is an evidence set of significant test results that may contain spurious evidence. The DIF evidence is partitioned into different subsets according to DIF sources and DIF items: $\text{SOURCE}(Y_i)$ is the subset of exogenous variables that appear to

be sources of DIF for Y_i , and $\text{DIF}(X_j)$ is the subset of items that appear to have DIF relative to X_j .

Step 3 attempts to straighten things out in three different ways: (a) stepwise elimination of LD evidence, (b) stepwise elimination of exogenous variables from $\text{SOURCE}(Y_i)$: $X_j \in \text{SOURCE}(Y_i)$ is removed if the hypothesis $Y_i \perp X_j | S, \text{SOURCE}(Y_i) \setminus X_j$ is accepted, (c) stepwise elimination of items from $\text{DIF}(X_j)$: $Y_i \in \text{DIF}(X_j)$ is removed if the hypothesis $Y_i \perp X_j | S, \text{DIF}(X_j) \setminus Y_i$ is accepted. The order of (a), (b) and (c) is of no consequence since $\text{SOURCE}(Y_i)$ and $\text{DIF}(X_j)$ are not changed until all steps have been taken, but other strategies where steps are taken sequentially are also possible.

The hypotheses tested during step 3(b) are GMP hypotheses if a single item is a DIF item, and the hypotheses tested in 3(c) are GMP hypotheses if there is only one DIF source. The result after these steps have been taken can still contain spurious evidence. To check this, GMP hypotheses derived from the remaining evidence of LD and DIF are tested in Step 6.

Step 4 is a backwards elimination search for the exogenous variables on which the score depends. This addresses issues relating to the structural framework of the GLLRM, and also tests criterion validity. The result may have some effect on the global Markov properties of the model defined in the next step and therefore also on the final check of the model in Step 6 below.

Step 5 collects the remaining evidence of LD and DIF into a GLLRM and moralizes the marginal Rasch graph to facilitate identification of minimal GMP hypotheses.

Step 6 tests C5, C7, and C9 hypotheses and other GMP hypotheses derived by separation in the moralized marginal Rasch graph and modify the model according to the results.

6.1. Examples: Item Screening of Simulated Data

Example 5. Step 2 of the item screening rejects ten out of the 20 GMP hypotheses. Adjustment by the Benjamini–Hochberg procedure to control the false discovery rate at 5% suggests that four of these hypotheses were rejected due to Type I errors. The remaining evidence indicates LD for (A, B) and (A, E) and DIF for (B, X) , (E, X) , (A, Z) , and (B, Z) .

Step 3 does not change the assessment of LD, but eliminates two DIF terms, since $B \perp Z | S, A$ ($\gamma = 0.07$, $p = 0.064$) and $E \perp X | S, B$ ($\gamma = -0.07$, $p = 0.060$).

Step 4 confirms that S depends on both X and Z following which Step 5 defines a GLLRM with two instances of LD and two instances of DIF. The check of the C7 and C9 hypotheses discards the evidence of LD,

$$\begin{aligned} A \perp B | R_B, Z : \gamma = 0.03, \quad p = 0.18, \quad A \perp B | R_{AE}, X : \gamma = 0.06, \quad p = 0.06, \\ A \perp E | R_E, X, Z : \gamma = -0.06, \quad p = 0.09, \quad A \perp E | R_{AB} : \gamma = -0.03, \quad p = 0.29. \end{aligned}$$

But the tests of the C5 hypotheses reject both hypotheses of no DIF. The end result is therefore a GLLRM with the AZ interaction in addition to the true BX interaction. Since the C5 test that rejects the hypothesis that the AZ interaction vanishes is a GMP hypothesis under the true model, we know that this result is a Type I error.

Example 6. In this example there was evidence of seven instances of LD. Step 3a assumes the LD result with the largest partial γ as true and discards results pertaining to C4 hypotheses that are not true if this assumption is correct. This is then repeated with the remaining evidence of LD. In this example the first step selects the BC interaction at the first step and discards all the other test results. Step 6 finds no additional evidence of LD, and the item screening thus finds the true model.

Example 7. The results obtained during item screening are extensive and are not shown here. In summary, all evidence in Table 4 survives the Benjamini–Hochberg adjustment. Step 3a selects

TABLE 6.

Partial γ coefficients for analysis of LD by tests of C4 hypotheses of conditional independence of items given a rest-score without the item defining the row, and for analysis of DIF by tests of C2 hypotheses of conditional independence of items and exogenous variables given the total score on all items. Test results regarded as significant after adjustment for multiple testing are written in **bold**.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>
<i>A</i> : Vigorous activities		0.40	0.10	−0.12	−0.24	−0.11	0.01	− 0.37	− 0.64	−0.25
<i>B</i> : Moderate activities	0.26		0.67	− 0.48	− 0.31	− 0.31	− 0.29	− 0.47	− 0.55	−0.02
<i>C</i> : Carrying groceries	−0.10	0.71		− 0.26	0.07	− 0.25	− 0.40	−0.33	−0.16	0.19
<i>D</i> : Climbing several flights of stairs	−0.18	− 0.35	−0.20		0.77	0.19	0.30	−0.12	−0.16	−0.33
<i>E</i> : Climbing one flight of stairs	− 0.49	− 0.45	−0.08	0.74		0.19	−0.27	0.07	0.22	0.00
<i>F</i> : Bending, kneeling, or stooping	−0.22	−0.15	−0.14	0.22	0.36		0.14	0.05	0.01	0.24
<i>G</i> : Walking more than a mile	−0.11	− 0.22	− 0.39	0.26	−0.15	0.04		0.68	0.36	−0.14
<i>H</i> : Walking several blocks	− 0.62	− 0.55	− 0.42	−0.21	0.06	−0.12	0.64		0.97	0.09
<i>I</i> : Walking one block	− 0.79	− 0.63	−0.29	−0.20	0.24	−0.17	0.29	0.97		0.31
<i>J</i> : Bathing or dressing yourself	−0.33	−0.08	0.15	−0.29	0.16	0.17	−0.13	0.24	0.43	
<i>K</i> : Health	−0.03	−0.05	0.26	−0.04	0.31	−0.02	−0.05	−0.01	−0.26	−0.24
<i>L</i> : BMI	0.05	0.31	0.21	−0.14	−0.20	− 0.17	−0.02	−0.12	0.04	−0.16
<i>M</i> : Smoking	−0.06	0.14	−0.07	0.23	−0.05	− 0.21	0.04	0.00	−0.16	0.27
<i>N</i> : Sex	0.23	− 0.45	− 0.45	−0.06	−0.09	0.14	0.02	0.20	−0.11	0.57
<i>O</i> : Age	−0.09	0.12	−0.17	0.07	− 0.24	0.14	−0.07	−0.02	0.05	0.19

the interactions AD, BD, CD and DE and discards remaining LD evidence. Steps 3b and 3c accept DIF of *D* relative to *X* and *Z* and discards remaining DIF evidence. The final check of C5, C7 and C9 hypotheses changes nothing. The end result is a GLLRM, where *X* and *Z* are DIF sources of *D* and where *D* shows LD with all other items. It is obvious that something is wrong with item *D*, but item screening fails to identify the problem. This illustrates that item screening targets a specific type of model and fails if the true model is a different kind of IRT model.

7. Item Screening of the Physical Functioning Subscale of SF36

This example uses data from a health study in Copenhagen in 1995 where 2546 persons responded to the ten items of the PF subscale of SF36 (Fayers & Machin, 2007) measuring physical functioning. Following the question “The following items are about activities you might do during a typical day. Does your health now limit you in these activities? If so, how much?”, the questionnaire lists ten different activities to which the respondent has to choose one of the following three response options: “Yes, limited a lot”, “Yes, limited a little”, and “No, not limited at all”. We include five exogenous variables in the analysis. The ten items labeled *A* to *J* and the exogenous variables labeled *K* to *O* can be seen in Table 6. Where convenient these labels will be used when we refer to the variables.

7.1. Item Screening Step 1: Analysis of Consistency

The analysis of two-way associations among items, the total score, and exogenous variables does not disclose evidence against consistency: items are marginally related to each other and to

TABLE 7.
Accepted evidence of local dependence.

Locally dependent items		Mean γ
<i>H</i> : Walking 2+ blocks	<i>I</i> : Walking 1 block	0.97
<i>D</i> : Stairs 2+ flights	<i>E</i> : One flight of stairs	0.76
<i>B</i> : Moderate activities	<i>C</i> : Carrying groceries	0.69
<i>G</i> : Walking one mile	<i>H</i> : Walking 2+ blocks	0.66
<i>A</i> : Vigorous activities	<i>B</i> : Moderate activities	0.33
<i>I</i> : Walking 1 block	<i>J</i> : Bathing	0.37
<i>E</i> : One flight of stairs	<i>F</i> : Bending	0.28

the score, and associations between the score and exogenous variables are reflected in the items. Evidence of criterion validity was seen as an association between the score and self-reported health ($\gamma = -0.61$, $p < 0.0005$).

7.2. Item Screening Step 2: Tests of GMP Hypotheses Derived from a Graphical Rasch Model

Table 6 shows tests of the GMP hypotheses of local independence and no DIF under the graphical Rasch model. The top half of Table 6 shows the partial γ coefficients measuring the association among items for the tests of C4 hypotheses. The lower part shows the partial γ coefficients measuring the association among items and exogenous variables for tests of C2 hypotheses. 47 LD tests and 14 DIF tests are significant at a 5% level. The Benjamini–Hochberg procedure controlling the false discovery rate at 5% accepts tests with $p > 0.01714$, resulting in 35 LD tests and 12 DIF tests remaining significant.

7.3. Item Screening Step 3a: Spurious Evidence of LD

During each step of the attempt to separate spurious from genuine evidence of LD, the item screening procedure takes the strongest remaining evidence of LD at face value and disregards hypotheses that are then known to be false. The strongest evidence point at LD between *H* and *I*. This is taken at face value and results obtained by conditioning on rest-scores R_H and R_I (the rows for *H* and *I* in Table 6) are ignored: evidence against $C \perp H | R_H$ ($\gamma = -0.42$, $p = 0.003$) is therefore disregarded. The next step assumes LD between *D* and *E* and discards evidence of LD between *A* and *E*. At the end, seven pairs of LD items are recognized (Table 7).

7.4. Item Screening Steps 3b and 3c: Spurious evidence of DIF

In Table 6, $\text{DIF}(N) = \{A, B, C, J\}$ and $\text{Source}(C) = \{K, L, N\}$. The question is whether the evidence that *N* is a DIF source for *C* is spurious. This is investigated in Table 8.

The left part of Table 8 examines evidence of *N* as a DIF source by tests of conditional independence of items in $\text{DIF}(N)$ and *N* given the score and other items of $\text{DIF}(N)$. Evidence that *A* has DIF relative to *N* is regarded as spurious ($\gamma = 0.14$, $p = 0.210$). To check that this does not alter conclusions for other items, the analysis is repeated. The remaining evidence of DIF relative to *N* appears to be genuine.

The right part of Table 8 examines whether *C* has three different DIF sources by tests of conditional independence of *C* and variables in $\text{Source}(C)$ conditional on the score and other DIF sources. Conditional independence of *C* and *N* is accepted ($\gamma = -0.17$, $p = 0.364$).

The results of this analysis and the analyses of the remaining evidence of DIF in Table 6 are summarized in Table 9. Five out of 12 cases of DIF evidence are regarded as spurious.

TABLE 8.

Assessment of the evidence of DIF for “Carrying groceries” (C) and Sex (N). The other variables considered are A = “Vigorous activities”, B = “Moderate activities”, J = “Bathing”, K = Self-reported health, L = BMI.

Hypothesis	Partial γ	p	Hypothesis	Partial γ	p
$A \perp N S, B, C, J$	0.14	0.210	$C \perp K S, L, N$	0.23	0.037
$B \perp N S, A, C, J$	−0.45	0.001	$C \perp L S, K, N$	0.19	0.020
$C \perp N S, A, B, J$	−0.39	0.004	$C \perp N S, K, L$	−0.17	0.364
$J \perp N S, A, B, C$	0.41	0.012			
$B \perp N S, C, J$	−0.49	0.001	$C \perp K S, L$	0.24	0.023
$C \perp N S, B, J$	−0.50	0.001	$C \perp L S, K$	0.21	0.005
$J \perp N S, B, C$	0.58	0.007			

TABLE 9.

Assessment of evidence of DIF. γ_1 is the partial γ coefficient under control for multiple items with DIF relative to the same DIF source, while γ_2 is the partial γ coefficient under control for multiple DIF sources.

DIF source	Item	γ_1	p	γ_2	p	Conclusion
K : SRH	C : Carrying groceries	0.30	0.004	0.24	0.023	DIF
	E : One flight of stairs	0.34	0.002	0.30	0.006	DIF
L : BMI	B : Moderate activities	0.29	0.001	0.23	0.002	DIF
	C : Carrying groceries	0.20	0.032	0.21	0.005	DIF
M : Smoking	F : Bending	−0.08	0.667			Spurious
	D : Stairs 2+ flights	0.23	0.012			DIF
	F : Bending	−0.19	0.071			Spurious
N : Sex	A : Vigorous activities	0.11	0.710			Spurious
	B : Moderate activities	−0.49	0.001	−0.43	0.002	DIF
	C : Carrying groceries	−0.50	0.001	−0.17	0.364	Spurious
	J : Bathing	0.58	0.007			DIF
O : Age	E : One flight of stairs			−0.21	0.054	Spurious

7.5. Item Screening Step 4: Association Between the Score and Exogenous Covariates

Step 4 considers the conditional association between the score and exogenous covariates conditionally given all other covariates. Conditional independence is rejected for all covariates (results not shown). The PF scale satisfies the requirement of strong criterion validity since the partial γ measuring the conditional association between the score and Health given Sex, Age, BMI, and Smoking is highly significant ($\gamma = -0.56$, $p < 0.0005$).

7.6. Item Screening Step 5: The GLLRM Model

Step 5 creates a GLLRM based on the evidence of DIF and LD that survived Step 3 and on the evidence of conditional dependence of the score on exogenous variables. Information on conditional independence among exogenous variables based on model search among these variables can also be incorporated in the model. The screening procedure of Kreiner (1986) may be helpful here. Figure 6a shows the IRT graph of this model. The role of this figure is, first of all, to give a visual summary of the status of the item screening analysis after Step 4. The main contribution of Step 5 is, however, the moralized marginal graph shown in Figure 6b, since this graph will serve as the starting point for the final part of the item screening.

7.7. Item Screening Step 6: Checking the GLLRM

Step 6 consists of three parts. First, C5, C7 and C9 tests check that the evidence supporting the GLLRM model is genuine. These tests dismiss three instances of LD and four instances of

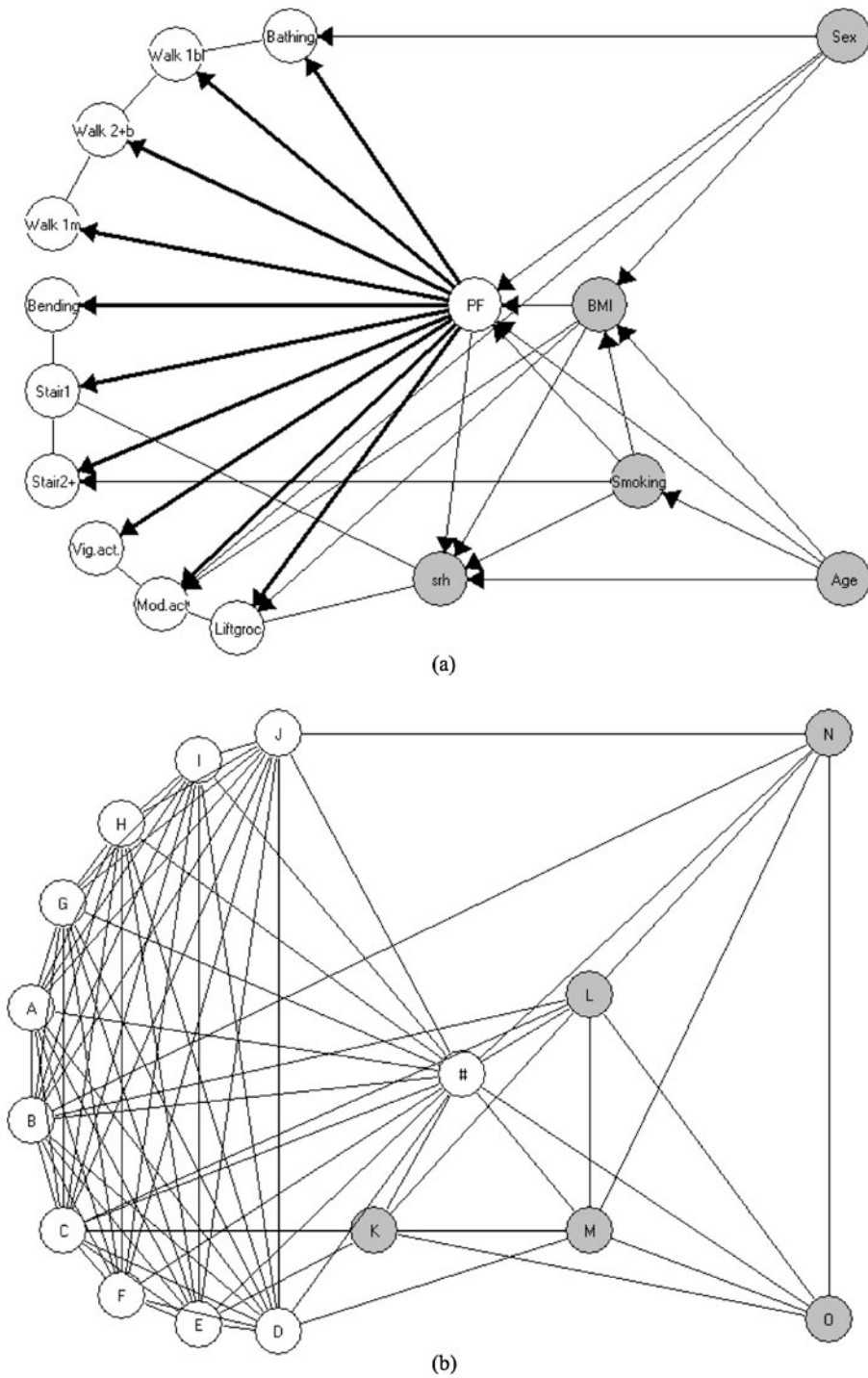


FIGURE 6.

The GLLRM defined at Step 5 during the screening of PF items. (a): The IRT graph. The PF variable in this figure is the latent physical functioning variable. (b): The moralized marginal Rasch graph final GLLRM defined by item screening. Variable names have been replaced by labels. The “#” symbol refers to the total score on all items.

TABLE 10.
Test of C7 and C9 hypotheses of local independence in the GLLRM defined in Figure 6.

Locally dependent items		C7: γ_1	C9: γ_2	Conclusion
<i>H</i> : Walking 2+ blocks	<i>I</i> : Walking 1 block	1.00 (<0.001)	0.95 (<0.001)	Local independence
<i>D</i> : Stairs 2+ flights	<i>E</i> : One flight of stairs	0.83 (<0.001)	0.69 (<0.001)	
<i>B</i> : Moderate activities	<i>C</i> : Carrying groceries	0.82 (<0.001)	0.74 (<0.001)	
<i>G</i> : Walking one mile	<i>H</i> : Walking 2+ blocks	0.31 (0.167)	0.39 (0.100)	
<i>A</i> : Vigorous activities	<i>B</i> : Moderate activities	0.21 (0.154)	0.49 (<0.001)	Local independence
<i>I</i> : Walking 1 block	<i>J</i> : Bathing	0.64 (0.054)	0.64 (0.040)	
<i>E</i> : One flight of stairs	<i>F</i> : Bending	0.36 (0.036)	0.40 (0.124)	Local independence

TABLE 11.
Test of C5 hypotheses of no DIF in the GLLRM defined in Figure 6.

DIF source	Item	γ	p	Conclusion
<i>K</i> : Health	<i>C</i> : Carrying groceries	0.43	0.017	DIF
	<i>E</i> : One flight of stairs	0.30	0.188	No DIF
<i>L</i> : BMI	<i>B</i> : Moderate activities	0.12	0.304	No DIF
	<i>C</i> : Carrying groceries	0.19	0.200	No DIF
<i>M</i> : Smoking	<i>D</i> : Stairs 2+ flights	0.35	0.004	DIF
<i>N</i> : Sex	<i>B</i> : Moderate activities	−0.72	0.001	DIF
	<i>J</i> : Bathing	0.70	0.101	No DIF

DIF (Tables 10 and 11, respectively). Second, C5, C7 and C9 tests check the resulting GLLRM and show no evidence of problems (results not shown).

The C5, C7 and C9 hypotheses tested during these two parts are GMP hypotheses, but they may not be minimal. Tests of minimal GMP hypotheses may be more powerful because they have conditioning sets with fewer variables. For this reason, the final part reanalyzes the evidence of DIF and LD relationships that the tests of C5, C7 and C9 hypotheses indicated were spurious.

The difference between minimal GMP hypotheses and C5 hypotheses of no DIF can be seen when testing DIF of Item *C* relative to *N*. The C5 hypothesis relating to *C* and *N* is $C \perp N | S, \text{DIF}(N), \text{Source}(C)$ where the conditioning set contains seven variables because $\text{DIF}(N) = \{B, D, E, J\}$ and $\text{SOURCE}(C) = \{K, L\}$. Minimal GMP hypotheses require the smallest subsets of variables that separate *C* and *N* in the moralized graph (Figure 6b). Three subsets of variables define minimal GMP hypotheses: (S, B, D, J, K, L) , (S, B, J, K, L, M) and (S, B, J, L, M, O) . We expect tests with these sets of conditioning variables to have greater power. In this case, however, none of these tests provide significant evidence against the hypothesis of no DIF.

Tests of the C7 and C9 hypotheses of LD for *G* and *H* were insignificant (Table 10). The moralized graph can be used to identify three minimal GMP hypotheses: (i) $G \perp H | \text{SDKLN}$, (ii) $G \perp H | \text{SKLMN}$, and (iii) $G \perp H | \text{SKLNO}$. Two of these tests reject local independence and we claim that evidence supports LD between these two items.

Figure 7 shows the model after elimination of the DIF and LD terms that were supported by spurious evidence according to Step 6. It is this model that item screening suggests we should use as the starting point for a search for an adequate GLLRM.

8. Discussion

In health research and medicine, screening is a strategy used to identify the persons of a population with a specific disease. In most cases, guidelines for screening require (1) that diagnostic

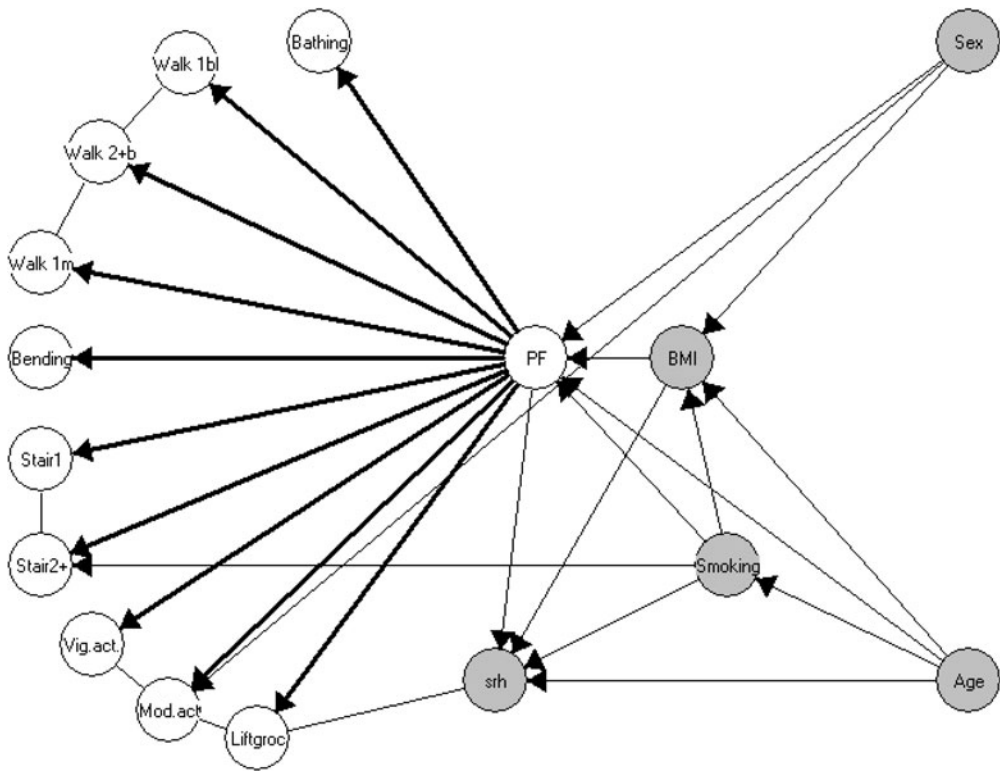


FIGURE 7.
The GLLRM defined at the end of Step 6 of the by item screening procedure.

tests used during screening are cheap, easy to perform and widely available, (2) that diagnostic tests are specific, sensitive and reliable with good positive and predictive values, and (3) that clinical diagnosis and treatment for the disease are available. In this paper we have developed and illustrated a screening procedure meeting the same requirements for initial item analysis of items for Rasch models.

8.1. The Diagnosis

During the item screening, the evidence of DIF and LD is provided by partial γ coefficients measuring conditional association among ordinal and binary variables that are easy to perform, have well-known asymptotic properties and where Monte Carlo assessment of significance available. The item screening procedure attempts to ‘diagnose’ the problem by adjustment for multiple testing and identification of spurious evidence. The methodical use of the global Markov properties of GLLRMs, and in particular the final model check in Step 6 of the item screening procedure goes a long way toward making this possible.

The success of the item screening depends on several factors. We make no claim that partial γ coefficients provide the most powerful tests of DIF and LD or that the Benjamini–Hochberg procedure yields the best adjustment for multiple testing. Indeed, these can be substituted by other techniques without destroying the basic idea of item screening and the usefulness of the global Markov properties.

8.2. *The Treatment*

Item screening provides ‘treatment’ in the shape of a GLLRM that can serve as a convenient starting point for model search for a GLLRM by parametric methods. Other screening procedures have been discussed in the psychometric literature, but the majority of these provide treatment by amputation (purification) rather than treatment by modelling. To our knowledge, our procedure is also the first procedure of its kind that simultaneously assesses evidence of DIF and LD.

Graphical Rasch models and GLLRMs belong to a broader family of graphical models with latent variables. Other examples are discussed by Hagnaars (1998), Anderson and Böckenholt (2000), Humphreys and Titterton (2003), Bartolucci and Forcina (2005), Bartolucci (2007), Anderson and Yu (2007), and Rijmen, Vansteelandt, and De Boeck (2008). The preponderance of such models is either undirected models or models defined by directed acyclic graphs (DAGs) where manifest variables are conditionally independent given the latent variables.

The choice between GLLRM modelling of LD and DIF and purification is a choice between loss of specific objectivity (in the GLLRM) and lower reliability and inflated standard errors of measurement (in the shorter scale). Both solutions yield an IRT model where the total score is sufficient for the person parameter and in which person parameters can be estimated without confounding by the exogenous variables. Kreiner and Christensen (2006) and Kreiner (2007) claim that measurement by items from a GLLRM is essentially valid and objective and argue that the difference between essential validity and objectivity and criterion-related construct validity (Rosenbaum, 1989) and objectivity (Rasch, 1961/2006) is less critical than the loss of reliability due to purification in a relatively small set of items. In short tests, exogenous variables provide extra information and help increase/improve measurement of θ .

Example 4 illustrates that the item screening is not a miracle cure; the treatment provided by GLLRMs may fail miserably if the problems with the items are other than LD and DIF problems.

8.3. *Usefulness?*

The item screening strategy described in this paper is implemented in DIGRAM (Kreiner, 2003), but may in principle be performed using standard statistical software. Item screening is particularly useful in situations with a complex DIF and LD structure, for example, in health related and sociological research with few items and obvious LD (cf. the SF36 example), and less useful when items fit the Rasch model since parametric analysis yields the same result as quickly as the item screening. An automatic item screening procedure like the one implemented in DIGRAM is so fast and provides so much information on consistency and criterion validity that we have also found it to be useful when the DIF and LD structure is simple or even non-existing since the analyses in Steps 1, 2 and 4 of consistency, LD, DIF and criterion validity are analyses that, to us, are routine components of analyses of fit of item response models anyway. If items are few and/or there is little DIF and little LD, the remaining parts of the item screening, Steps 3, 5 and 6, are close to instantaneous.

Acknowledgements

We would like to thank Mounir Mesbah, the editors of Psychometrika and three anonymous reviewers for their careful reading and helpful comments and suggestions.

Appendix A. Chain Graph Models

The vocabulary of chain graph models is extensive. We need to remind the reader about some of the notions of parents, paths, separation, chain components and moralized graphs from the theory of chain graph models for the results on global Markov properties to make sense.

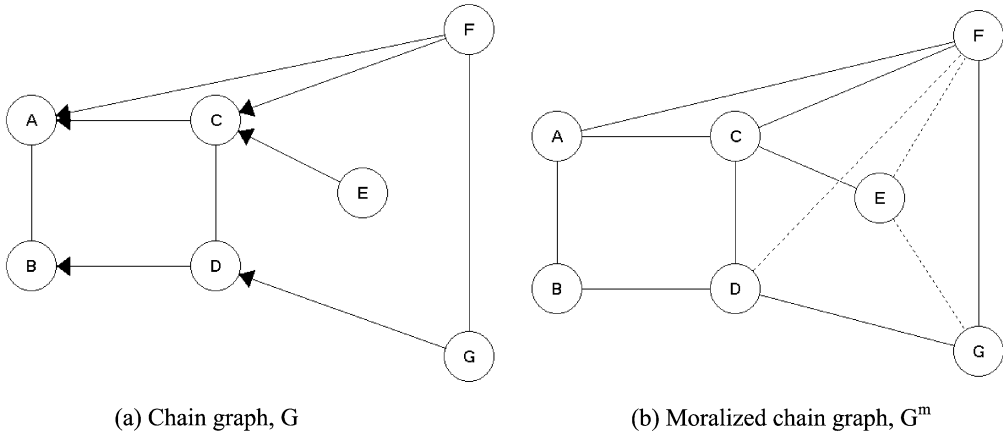


FIGURE 8.

A Chain graph with the corresponding moralized graph. The *dashed edges* in the moral graph have been added during moralization.

A mathematical graph is a pair $G = (V, E)$, where V is a finite set of nodes and $E \subseteq V \times V$ represents links between nodes. Graphs are often displayed as visual networks with dots representing nodes and edges or arrows representing links between nodes. Two nodes A and B are joined by an undirected edge in the visual graph if $(A, B) \in E$ and $(B, A) \in E$ or by a directed edge (an arrow) pointing from A to B if $(A, B) \in E$ and $(B, A) \notin E$. A subset $W \subset V$ defines a subgraph (W, F) where $F = E \cap (W \times W)$. When it is obvious that W is a subset of nodes of a graph G we refer to the subgraph as G_W .

Chain graphs: are graphs where the set of nodes is partitioned into a numbered sequence of disjoint node blocks where connected nodes in the same block are joined by undirected edges while connected nodes from different blocks are joined by arrows pointing from blocks with high numbers toward blocks with lower numbers. Figure 8(a) is a chain graph.

Undirected graphs: a chain graph with but one recursive block is referred to as an undirected graph since it only contains undirected edges. The graph in Figure 8(b) is undirected.

DAGs: chain graphs with only one node in each block are called *directed acyclic graphs*.

Parents: let A and B be two nodes in a chain graph. B is called a parent of A if the graph contains an arrow pointing from B to A . Nodes can have several parents in the same or different recursive blocks. The set of parents of A are called $\text{pa}(A)$. In Figure 8(a), C and F are parents of A .

Paths: a path between two nodes in a chain graph is a sequence of attached edges between the two nodes. The path may contain both arrows and undirected edges. If arrows are included, they must all point in the same direction. In Figure 8(a), $G \rightarrow D \rightarrow B \rightarrow A$ is a path.

Separation: let A and B be disjoint subsets of nodes of a chain graph. We say that a subset of nodes, C , separates A from B if all paths between nodes of A and nodes of B pass through or are intersected by a node of C . In Figure 8(a), $\{A, D\}$ separates B from G .

Chain components: a chain component is a maximal subset of nodes in a recursive block where all pairs of nodes are connected by at least one path in the block. If there is no path between two variables in the same block they must belong to different chain components. If $\{E, F, G\}$ belong to the same block in Figure 8(a), this block has two chain components $\{E\}$ and $\{F, G\}$.

Boundaries: the boundary of a set of nodes A consist of all nodes that are connected to at least one node in A by either an undirected edge or by an arrow pointing into A . $\{A, B, C, D, G\}$ is the boundary of $\{B, D\}$ in Figure 8(a).

Borders: let $\text{bd}(A)$ be the boundary of a set of nodes. Since $\text{bd}(A)$ can contain some of the nodes of A , we define the border of A as $\text{bd}(A) \setminus A$. The border of $\{B, D\}$ in Figure 8(a) is $\{A, C, G\}$.

Ancestral sets: a set of nodes is ancestral if the border is empty. For any subset of nodes, A , there exists a smallest ancestral set, $\text{an}(A)$, containing A . The graph in Figure 8(a) defines five ancestral sets $\{A, B, C, D, E, F, G\}$, $\{C, D, E, F, G\}$, $\{E, F, G\}$, $\{E\}$, $\{F, G\}$.

Moral graphs: the moral graph G^M associated with G is an *undirected* graph containing the same nodes and undirected edges as G , where arrows in G have been replaced by undirected edges, and where undirected edges are added between nodes that are parents to nodes in the same chain components of G . Figure 8(a) shows the Markov graph of a model containing three recursive blocks, $[A, B] \leftarrow [C, D] \leftarrow [E, F, G]$. The moral graph associated with the chain graph is shown in Figure 8(b). In this graph, an undirected edge between D and F has been added, since D is a parent of B , F is a parent of A , and A and B are connected in Figure 8(a).

(GMP) Global Markov properties: of the many results from the theory of chain graph models, the fact that conditional independence in marginal models may be read directly off the moral graphs is particularly useful. This follows from the global Markov properties (Lauritzen, 1996, p. 55).

Assume that (A, B, C) are disjoint subsets of nodes of a Markov graph and let $D = \text{an}(A \cup B \cup C)$. Next let G_D be the subgraph of D and let G_D^m be the moralized subgraph. If C separates A and B in G_D^m then $A \perp B | C$.

Since G in itself is ancestral, it follows that $A \perp B | C$ if C separates A and B in G^m . Since A and D are separated by $\{B, C, F\}$ in G^m , it follows that $A \perp D | B, C, F$.

(LMP) Local Markov properties: The local chain Markov properties apply if $A \perp B | \text{bd}(A)$ for all pairs of variables, (A, B) , where $B \not\subset \text{bd}(A)$.

(PMP) Pairwise Markov properties: The conditional independence assumptions that define a chain graph model are called pairwise Markov properties. Lauritzen (1996, Theorem 3.34) shows that the pairwise Markov properties and the global Markov properties are equivalent if the joint probabilities of all variables are positive.

If the probabilities and densities of all possible combinations of outcomes on the variables of the model are positive, it follows from Theorem 3.34 of Lauritzen (1996) that $(\text{GMP}) \Leftrightarrow (\text{LMP}) \Leftrightarrow (\text{PMP})$.

GMP hypotheses: Kreiner, Pedersen, and Siersma (2009) refer to the conditional independence statements defined by the global Markov properties as GMP hypotheses. The hypothesis, $A \perp D | B, C, F$, derived from Figure 8 is a GMP hypothesis.

Minimal GMP hypotheses: A chain graph is by definition ancestral, but it does not have to be the smallest ancestral set containing A , B and C . Also, let A, B, C_1, C_2 be disjoint subsets of nodes of a chain graph G with moralized graph equal to G^m . If C_1 separates A and B in G^m it follows that $G_1 \cup G_2$ also separates A and B and therefore that $A \perp B | C_1, C_2$. For this reason, Kreiner et al. (2009) say that a GMP hypothesis $A \perp B | C$ is *minimal* if there are no smaller subsets, $D \subset C$, that separate A and B in the ancestral set, $\text{an}(A \cup B \cup D)$.

A.1. Proof of Theorem 1 on functional collapsibility in chain graph models

Let $\mathbf{X}$ and $\mathbf{Y}$ be vectors of random variables and assume that $P(\mathbf{X}, \mathbf{Y})$ is a chain graph model with Markov graph G_{XY} . Also, let S be function of $\mathbf{Y}$, $S = f(\mathbf{Y})$. We say that the chain graph model is functionally collapsible onto $(\mathbf{X}, S)$ if $\mathbf{X} \perp \mathbf{Y} | S$.

Lauritzen (1996, p. 29) refers to two results concerning conditional independence involving functions of variables that are useful in a discussion of the implications of functional collapsibility. They are

- (a) if $\mathbf{X} \perp \mathbf{Y} | \mathbf{Z}$ and $\mathbf{U} = f(\mathbf{X})$ then $\mathbf{U} \perp \mathbf{Y} | \mathbf{Z}$,

(b) if $\mathbf{X} \perp \mathbf{Y} | \mathbf{Z}$ and $\mathbf{U} = f(\mathbf{X})$ then $\mathbf{X} \perp \mathbf{Y} | \mathbf{Z}, \mathbf{U}$.

If $\mathbf{Y}$ and $\mathbf{X}$ are vectors, it follows functional collapsibility ($\mathbf{X} \perp \mathbf{Y} | S = f(\mathbf{Y})$) together with (a) that $X_i \perp X_j | S$ since Y_i is a function of Y and X_j is a function of $\mathbf{X}$.

To prove Theorem 1, we need the following lemma summarizing the consequences of functional collapsibility on the joint distributions of $(\mathbf{X}, S, \mathbf{Y})$ and $(S, \mathbf{X})$.

Lemma A.1. *Let $\mathbf{X}_A \subseteq \mathbf{X} \setminus \{X_i, X_j\}$. If $\mathbf{X} \perp \mathbf{Y} | S = f(\mathbf{Y})$ and $X_i \perp X_j | \{\mathbf{X}_A, \mathbf{Y}\}$ then $X_i \perp X_j | \{\mathbf{X}_A, S\}$.*

Proof: It follows from $X_i \perp X_j | \{\mathbf{X}_A, \mathbf{Y}\}$ that

$$\begin{aligned} P(X_i, X_j, \mathbf{X}_A, \mathbf{Y}) &= \frac{P(X_i, \mathbf{X}_A, \mathbf{Y}) \cdot P(X_j, \mathbf{X}_A, \mathbf{Y})}{P(\mathbf{X}_A, \mathbf{Y})} \\ &= \frac{P(X_i, \mathbf{X}_A, \mathbf{Y} | S) \cdot P(S) \cdot P(X_j, \mathbf{X}_A, \mathbf{Y} | S) \cdot P(S)}{P(\mathbf{X}_A, \mathbf{Y} | S) \cdot P(S)}. \end{aligned}$$

From functional collapsibility, $\mathbf{X} \perp \mathbf{Y} | S$, and the fact that $P(\mathbf{Y} | S)P(S) = P(\mathbf{Y}, S) = P(\mathbf{Y})$, since S is a function of $\mathbf{Y}$, it follows that

$$P(X_i, X_j, \mathbf{X}_A, \mathbf{Y}) = \frac{P(X_i, \mathbf{X}_A | S) \cdot P(\mathbf{Y}) \cdot P(X_j, \mathbf{X}_A | S) \cdot P(\mathbf{Y})}{P(\mathbf{X}_A | S) \cdot P(\mathbf{Y})}$$

and therefore first that

$$P(X_i, X_j, \mathbf{X}_A, \mathbf{Y}) = P(X_i | \mathbf{X}_A, S) P(X_j | \mathbf{X}_A, S) P(\mathbf{X}_A | S) P(\mathbf{Y})$$

and second that

$$\begin{aligned} P(X_i, X_j | \mathbf{X}_A, S) &= \frac{\sum_{\mathbf{Y}: F(\mathbf{Y})=S} (P(X_i | \mathbf{X}_A, S) \cdot P(X_j | \mathbf{X}_A, S) \cdot P(\mathbf{X}_A | S) P(\mathbf{Y}))}{P(\mathbf{X}_A, S)} \\ &= \frac{P(X_i | \mathbf{X}_A, S) \cdot P(X_j | \mathbf{X}_A, S) \cdot P(\mathbf{X}_A | S) \cdot P(S)}{P(\mathbf{X}_A, S)} \\ &= P(X_i | \mathbf{X}_A, S) \cdot P(X_j | \mathbf{X}_A, S) \end{aligned}$$

which was to be proven. $\square$

Lemma A.1 implies that $X_i \perp X_j | \mathbf{X}_A, S, \mathbf{Y}_B$, where $Y_B \subseteq \mathbf{Y}$ when $\mathbf{X} \perp \mathbf{Y} | S = f(\mathbf{Y})$ and $X_i \perp X_j | \mathbf{X}_A, \mathbf{Y}$. To see this, we just have to use Lemma A.1 together with (b) and the function $g(\mathbf{Y}) = (f(\mathbf{Y}), \mathbf{Y}_B)$.

Given Lemma A.1, the proof of Theorem 1 is as follows.

Proof of Theorem 1: Let G_{XY}^M and G_{XYS}^M be the moralized version of G_{XY} and G_{XYS} . To prove Theorem 1, we consider all minimal separating sets in G_{XYS}^M for variables that are not connected by edges in G_{XYS}^M and prove that these variables are conditionally independent given the separating set and arbitrary subsets of additional variables from the model. It is convenient to consider three different situations. The proof in the first case is straightforward. The second and third cases require some remarks on the difference between G_{XY}^M and G_{XYS}^M to make it apparent how the GMP hypotheses defined by G_{XYS}^M may be derived from the global Markov properties of G_{XY} . These remarks follow after the discussion of the first case.

TABLE 12.
The distribution of (X, Z) and the conditional means of Θ .

X	Z	$P(X, Z)$	$E(\Theta X, Z)$	$sd(\Theta X, Z)$
1	1	0.40	-1.0	1.0
2	1	0.10	0.0	1.0
1	2	0.10	0.5	1.0
2	2	0.40	1.5	1.0

(i) All pairs of variables (Y_i, X_j) are separated by S in the G_{XYS} and therefore also by (S, X_A, Y_B) where $\mathbf{Y}_A \subseteq \mathbf{Y} \setminus Y_i$ and $\mathbf{X}_B \subseteq \mathbf{X} \setminus X_j$. Again, according to Lauritzen (1996, p. 29), $\mathbf{X} \perp \mathbf{Y} | \mathbf{Z}$ and $\mathbf{U} = h(\mathbf{X})$ implies that $\mathbf{X} \perp \mathbf{Y} | (\mathbf{Z}, \mathbf{U})$. We apply this relationship twice to show that $\mathbf{Y} \perp \mathbf{X} | S, \mathbf{Y}_A, \mathbf{X}_B$ from which it follows that $Y_i \perp X_j | S, Y_A, X_B$.

Assume next that there is no edge between X_i and X_j in G_{XYS}^M , and that a minimal separating set for the two variables has been identified. Since $\mathbf{Y}$ is only connected to $\mathbf{X}$ through S , it follows that the separating set either consists of nothing but $\mathbf{X}$ variables, $\mathbf{X}_C \subseteq \mathbf{X} \setminus \{X_i, X_j\}$ or that it consists of S and a subset of $\mathbf{X}$ variables, $\{S, \mathbf{X}_C\}$. In both cases, $\mathbf{X}_C$ may of course be empty. The two different situations are treated as different cases below, even though the arguments are very similar.

To derive GMP hypotheses corresponding to separation properties of G_{XYS}^M from the global Markov property of G_{XY} , we must compare separation in G_{XYS}^M to the separation in G_{XY}^M . All variables belonging to $\text{pa}(S)$ in G_{XYS} are connected in G_{XYS}^M . If $\mathbf{Y}$ contains more than one chain component, G_{XY}^M may have fewer edges than G_{XYS}^M since edges generated by moralization only adds edges between variables that are parents to nodes in the same chain component. From this it follows that all paths in G_{XY}^M are also paths in G_{XYS}^M . If $\mathbf{X}_C$ separates X_i and X_j in G_{XYS}^M , $\mathbf{X}_C$ will also be a separating set in G_{XY}^M . This leads directly to the first of the two remaining cases.

(ii) Assume that (X_i, X_j) are separated by a minimal separating set, $\mathbf{X}_C \subseteq \mathbf{X} \setminus \{X_i, X_j\}$ in G_{XYS} , that $\mathbf{X}_A \subseteq \mathbf{X} \setminus \{X_i, X_j, \mathbf{X}_C\}$ and that $\mathbf{Y}_B \subseteq \mathbf{Y}$. From the arguments above, $\mathbf{X}_C$ separates (X_i, X_j) in G_{XY}^M . For this reason, $\mathbf{X}_C, \mathbf{X}_A, \mathbf{Y}$ also separate (X_i, X_j) in G_{XY}^M . It therefore follows from the global Markov properties that $X_i \perp X_j | \mathbf{X}_C, \mathbf{X}_A, \mathbf{Y}$. From this and Lemma A.1 it follows that $X_i \perp X_j | \mathbf{X}_C, \mathbf{X}_A, S, \mathbf{Y}_B$.

(iii) Assume finally that a minimal separating set for (X_i, X_j) in G_{XYS}^M contains S in addition to $\mathbf{X}_C$. Since S is included in a minimal separating set, there must be at least one path from X_j to X_i passing through S . This path contains two nodes directly connected by edges to S in G_{XYS}^M and therefore also, according to the definition of G_{XYS} , a subset of nodes, $\mathbf{Y}_D \subseteq \mathbf{Y}$, in G_{XY}^M . From this it follows that (X_i, X_j) are separated by $\{\mathbf{X}_C, \mathbf{Y}_D\}$ and therefore also by $\{\mathbf{X}_A, \mathbf{X}_C, \mathbf{Y}\}$ in G_{XY}^M . From this and Lemma A.1 it once again follows that $X_i \perp X_j | \mathbf{X}_C, \mathbf{X}_A, S, \mathbf{Y}_B$. $\square$

Appendix B. The Parameters of the Baseline Model for Examples 2–4

In Examples 2–4 in Section 5, A, B, C, D , and E are dichotomous items, X and Z are stochastically dependent exogenous variables and Θ is the latent variable. The distribution of Θ is normal. The mean depends on X and Z , but the standard deviation is equal to 1 in all groups. Sample size is 10,000 in all three examples. The item parameters of the baseline model are shown below, and the distribution of (X, Z) and the conditional means of Θ are shown in Table 12.

Item parameters				
A	B	C	D	E
1.50	0.75	0.00	-0.75	-1.50

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Manuscript Received: 17 APR 2009

Final Version Received: 1 OCT 2010

Published Online Date: 9 MAR 2011